

Theoretical Physics 2 - Blatt 03

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7. Moments of statistical distributions Compute the expectation value $\mu = \langle X \rangle$ and the variance $\sigma = \langle (X - \mu)^2 \rangle$ for the given distributions.

- (a) Bernoulli distribution with $P(X) = p^x(1-p)^{1-x}$, $X \in \{0,1\}$

Calculation

Since X only takes the values 0 and 1, we have $P(X=0) = 1-p$, $P(X=1) = p$.

Therefore $\mu = \langle X \rangle = \sum_{X \in \{0,1\}} X P(X) = 0 \cdot (1-p) + 1 \cdot p = p$.

For the variance, we use $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \sum_{X \in \{0,1\}} X^2 \cdot P(X) + p^2 = p + p^2 = p \cdot (1+p)$

- (b) Poisson distribution with $P_\lambda(X) = \frac{\lambda^X e^{-\lambda}}{X!}$, $X \in \mathbb{N}$

Calculation

We compute the expectation value: $\mu = \langle X \rangle = \sum_{X=0}^{\infty} X \frac{\lambda^X e^{-\lambda}}{X!}$. The term $X=0$ vanishes and so we can pull X into the factorial: $\mu = \sum_{X=1}^{\infty} \frac{\lambda^X e^{-\lambda}}{(X-1)!} = \lambda e^{-\lambda} \sum_{X=1}^{\infty} \frac{\lambda^{X-1}}{(X-1)!} =$

$[n := X-1] = \lambda e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = \lambda e^{-\lambda} e^\lambda = \lambda$.

for computing the variance, we use $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \langle X(X-1) + X \rangle - \langle X \rangle^2 = \langle X(X-1) \rangle + \langle X \rangle + \langle X \rangle^2$, $\langle X(X-1) \rangle = \sum_{X=0}^{\infty} X(X-1) \frac{\lambda^X e^{-\lambda}}{X!}$

Similar for terms $X=0$, $X-1=0$ can be ignored, so we get: $\langle X(X-1) \rangle = \lambda^2 e^{-\lambda} \sum_{X=2}^{\infty} \frac{\lambda^{X-2}}{(X-2)!} = \lambda^2$. Therefore $\sigma^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$

- (c) Furthermore, compute μ , σ^2 , as well as the skewness $\mu_3 = \langle (X - \mu)^3 \rangle$ for the Gaussian distribution with probability (density) $P(X) = \frac{1}{\sqrt{2\pi b}} e^{-\frac{(X-a)^2}{2b}}$, where $X, a, b \in \mathbb{R}$.

Calculation

This is the normal distribution with mean parameter a and variance parameter b .

Calculate $\mu = \langle X \rangle = \int_{-\infty}^{\infty} dX X P(X) = \int_{-\infty}^{\infty} dX X \frac{1}{\sqrt{2\pi b}} e^{-\frac{(X-a)^2}{2b}} =$

$$\begin{aligned}
&= [y := X - a, dy = dX] = \int_{-\infty}^{\infty} dy (y + a) \frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} = \\
&= a \int_{-\infty}^{\infty} dy \frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} + \int_{-\infty}^{\infty} dy y \frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} = \\
&= a \frac{\sqrt{2\pi b}}{\sqrt{2\pi b}} + \int_0^{\infty} dy y \frac{1}{\sqrt{2\pi b}} e^{-\frac{y^2}{2b}} - \int_0^{\infty} (-dy) (-y) \frac{1}{\sqrt{2\pi b}} e^{-\frac{(-y)^2}{2b}} = a
\end{aligned}$$

and compute the variance $\sigma^2 = \langle (X - \mu)^2 \rangle = \langle y^2 \rangle = \frac{1}{\sqrt{2\pi b}} \int_{-\infty}^{\infty} dy y^2 e^{-\frac{y^2}{2b}} =$

$$\frac{1}{\sqrt{2\pi b}} \int_{-\infty}^{\infty} dy y \cdot y e^{-\frac{y^2}{2b}} = \frac{1}{\sqrt{2\pi b}} \left[\left(y \cdot \int dy y e^{-\frac{y^2}{2b}} \right)_{-\infty}^{\infty} - \int_{-\infty}^{\infty} dy \int dy y e^{-\frac{y^2}{2b}} \right]$$

$$\int dy y e^{-\frac{y^2}{2b}} = [z = \frac{-y^2}{2b}, \frac{dz}{dy} = -\frac{y}{b}, dy y = dz(-b)] = -\int dz b e^z = -b e^z = -b e^{-\frac{y^2}{2b}}$$

Therefore $\sigma^2 = \frac{1}{\sqrt{2\pi b}} \int_{-\infty}^{\infty} dy y \cdot y e^{-\frac{y^2}{2b}} = \frac{1}{\sqrt{2\pi b}} \left[\left(y \cdot (-b e^{-\frac{y^2}{2b}}) \right)_{-\infty}^{\infty} + \int_{-\infty}^{\infty} dy b e^{-\frac{y^2}{2b}} \right] =$

$$\frac{1}{\sqrt{2\pi b}} [b \sqrt{2\pi b}] = b$$

Now computing the third moment $\langle (X - \mu)^3 \rangle = \langle y^3 \rangle = \frac{1}{\sqrt{2\pi b}} \int_{-\infty}^{\infty} dy y^3 e^{-\frac{y^2}{2b}} =$

$$= \frac{1}{\sqrt{2\pi b}} \left[\int_0^{\infty} dy y^3 e^{-\frac{y^2}{2b}} - \int_0^{\infty} (-dy) (-y)^3 e^{-\frac{(-y)^2}{2b}} \right] = 0$$

8. Charged 1D harmonic oscillator in a uniform electric field We consider a particle of mass m and charge e moving in a (1D) harmonic oscillator potential $V(X) = \frac{1}{2}m\omega^2 X^2$ and additionally in a potential $-e\mathcal{E}X$ generated by a constant electric field $\mathcal{E}$. The Hamiltonian is so given by $H(\mathcal{E}) = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 - e\mathcal{E}X$.

Compute the energy eigenvalues $E_n(\mathcal{E})$ and eigenfunctions $\phi_n(x)$. Discuss the energy shift and the wave functions, and in particular compute the expectation value of the electric dipole moment $\langle D \rangle = e\langle \phi_n, X \phi_n \rangle$.

Hint: Complete the potential to a square in x and introduce the new variable $u = x - \frac{e\mathcal{E}}{m\omega^2}$. Show that u and P still satisfy the canonical commutation relations, i.e., they remain a pair of canonically conjugate variables.

Calculation

Compute Energy spectrum, shifted coordinate, eigenfunctions and dipole moment.

$$\frac{1}{2}m\omega^2 x^2 - eEx = \frac{1}{2}m\omega^2 \left(x^2 - 2\frac{eE}{m\omega^2}x \right) = \frac{1}{2} \left(x - \frac{eE}{m\omega^2} \right)^2 - \frac{1}{2}m\omega^2 \left(\frac{eE}{m\omega^2} \right)^2 = \frac{1}{2} \left(x - \frac{eE}{m\omega^2} \right)^2 - \frac{e^2 E^2}{2m\omega^2}$$

substitution: $u := x - \frac{eE}{m\omega^2}$

$$H(E) = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 u^2 - \frac{e^2 E^2}{2m\omega^2}$$

Since u differs by a constant: $[u, P] = [x - \frac{eE}{m\omega^2}, P] = [x, P] - [\frac{eE}{m\omega^2}, P] = [x, P] - 0 = i\hbar$.

So u, P are canonically conjugate variables.

For the normal harmonic oscillator the eigenvalues are $E_n^{(0)} \hbar\omega(n + \frac{1}{2})$, $n \in \mathbb{N}_0$ so for our case the eigenvalues are $E_n(E) = \hbar\omega(n + \frac{1}{2} - \frac{e^2 E^2}{2m\omega^2})$

usual eigenfunctions centered at $x_0 = \frac{eE}{m\omega^2}$, so $\phi_n(x; E) = \phi_n(x - \frac{eE}{m\omega^2})$, where ϕ_n is the n -th eigenfunction of the standard harmonic oscillator:

$$\phi_n(u) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} H_n \left(\sqrt{\frac{m\omega}{\hbar}} u \right) e^{-\frac{m\omega}{2\hbar} u^2}.$$

$$\text{Therefore } \phi_n(x; E) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} H_n \left(\sqrt{\frac{m\omega}{\hbar}} \left(x - \frac{eE}{m\omega^2} \right) \right) e^{-\frac{m\omega}{2\hbar} \left(x - \frac{eE}{m\omega^2} \right)^2}.$$

For the electric dipole moment we use $\langle D \rangle_n = e \langle X \rangle_n$. Since $X = u + \frac{eE}{m\omega^2}$ and the harmonic oscillator eigenstates satisfy $\langle u \rangle_n = 0$, we get $\langle X \rangle_n = \langle u + \frac{eE}{m\omega^2} \rangle_n = \langle u \rangle_n + \frac{eE}{m\omega^2} = \frac{eE}{m\omega^2}$.

$$\text{Therefore } \langle D \rangle_n = e \langle X \rangle_n = \frac{e^2 E}{m\omega^2}.$$

9. Coherent states of the 1D harmonic oscillator: properties A particle of mass m moves in a 1D harmonic oscillator potential with angular frequency ω . We define coherent (or “quasi-classical”) states by $\phi_\alpha := e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \phi_n$, ($\alpha \in \mathbb{C}$).

Here, ϕ_n are the energy eigenstates, i.e. $\hat{H}\phi_n = \hbar\omega(n + \frac{1}{2})\phi_n$, of the Hamiltonian $\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + \frac{1}{2})$.

(a) Show that coherent states are eigenstates of the annihilation operator: $\hat{a}\phi_\alpha = \alpha\phi_\alpha$.

Calculation

$$\text{We use } \hat{a}\phi_n = \sqrt{n}\phi_{n-1}, \quad \hat{a}\phi_0 = 0.$$

$$\text{Then } \hat{a}\phi_\alpha = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \hat{a}\phi_n$$

$$= e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sqrt{n} \phi_{n-1}$$

$$= e^{-|\alpha|^2/2} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{(n-1)!}} \phi_{n-1}. \text{ Set } m = n - 1: \hat{a}\phi_\alpha = e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^{m+1}}{\sqrt{m!}} \phi_m$$

$$= \alpha e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^m}{\sqrt{m!}} \phi_m$$

$$= \alpha\phi_\alpha.$$

(b) Show that coherent states are normalized: $\|\phi_\alpha\| = 1$.

Solution

$$\text{Using orthonormality } (\phi_n, \phi_m) = \delta_{nm}, \|\phi_\alpha\|^2 = (\phi_\alpha, \phi_\alpha) =$$

$$= e^{-|\alpha|^2} \sum_{n,m=0}^{\infty} \frac{(\alpha^*)^m \alpha^n}{\sqrt{m!n!}} (\phi_m, \phi_n) = e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = e^{-|\alpha|^2} e^{|\alpha|^2} = 1.$$

(c) Show completeness: $\frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha \phi_\alpha(\phi_\alpha, \psi) = \psi \quad \forall \psi \in L^2(\mathbb{R})$.

Calculation

$$\text{It is sufficient to show that the operator } \hat{I}_c := \frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha |\phi_\alpha\rangle\langle\phi_\alpha|$$

acts as the identity on the number basis $\{\phi_n\}$. Insert the coherent-state expansion:

$$|\phi_\alpha\rangle\langle\phi_\alpha| = e^{-|\alpha|^2} \sum_{n,m=0}^{\infty} \frac{\alpha^n (\alpha^*)^m}{\sqrt{n!m!}} |\phi_n\rangle\langle\phi_m|.$$

$$\text{Therefore } \hat{I}_c = \frac{1}{\pi} \sum_{n,m=0}^{\infty} \frac{|\phi_n\rangle\langle\phi_m|}{\sqrt{n!m!}} \int_{\mathbb{C}} d^2\alpha e^{-|\alpha|^2} \alpha^n (\alpha^*)^m.$$

Using polar coordinates $\alpha = re^{i\theta}$, one gets the standard integral

$$\frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha e^{-|\alpha|^2} \alpha^n (\alpha^*)^m = n! \delta_{nm}.$$

$$\text{So } \hat{I}_c = \sum_{n,m=0}^{\infty} \frac{n! \delta_{nm}}{\sqrt{n!m!}} |\phi_n\rangle\langle\phi_m| = \sum_{n=0}^{\infty} |\phi_n\rangle\langle\phi_n| = [1].$$

$$\text{Therefore, } \frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha \phi_\alpha(\phi_\alpha, \psi) = \psi$$

- (d) Show that the probability to measure the energy $E_n = \hbar\omega \left(n + \frac{1}{2}\right)$ in the state ϕ_α is Poisson distributed.

Solution

The amplitude for the level n is $\langle \phi_n | \phi_\alpha \rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$. Therefore the probability is $P_n = |\langle \phi_n | \phi_\alpha \rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$. This is precisely a Poisson distribution with parameter $\lambda = |\alpha|^2$.

- (e) Show that the mean energy and energy uncertainty are $\langle H \rangle_\alpha = \hbar\omega \left(|\alpha|^2 + \frac{1}{2}\right)$, $\Delta H_\alpha = \hbar\omega |\alpha|$.

Calculation

Since $\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2}\right)$, $\hat{N} = \hat{a}^\dagger \hat{a}$, it is enough to compute the moments of $\hat{N}$.

Because $\hat{a}\phi_\alpha = \alpha\phi_\alpha$, we get $\langle \hat{N} \rangle_\alpha = \langle \phi_\alpha | \hat{a}^\dagger \hat{a} | \phi_\alpha \rangle = \langle \hat{a}\phi_\alpha | \hat{a}\phi_\alpha \rangle = |\alpha|^2$.

So $\langle H \rangle_\alpha = \hbar\omega \left(\langle \hat{N} \rangle_\alpha + \frac{1}{2}\right) = \hbar\omega \left(|\alpha|^2 + \frac{1}{2}\right)$.

Next, $\Delta H_\alpha = \hbar\omega \Delta N_\alpha$. So we compute $\Delta N_\alpha^2 = \langle \hat{N}^2 \rangle_\alpha - \langle \hat{N} \rangle_\alpha^2$. Since the number distribution is Poisson with parameter $|\alpha|^2$, its variance is also $|\alpha|^2$. Therefore $\Delta N_\alpha^2 = |\alpha|^2 \Rightarrow \Delta N_\alpha = |\alpha|$. So $\Delta H_\alpha = \hbar\omega |\alpha|$.

The relative energy fluctuation behaves as $\frac{\Delta H_\alpha}{\langle H \rangle_\alpha} = \frac{|\alpha|}{|\alpha|^2 + \frac{1}{2}} \sim \frac{1}{|\alpha|}$ ($|\alpha| \gg 1$), so it vanishes in the quasi-classical limit.

- (f) Show that the position and momentum expectation values and uncertainties are

$$\langle X \rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}} \Re(\alpha), \quad \langle P \rangle_\alpha = \sqrt{2m\hbar\omega} \Im(\alpha), \text{ and } \Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}, \quad \Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}.$$

Calculation

We use the operator identities $X = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger)$, $P = -i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a} - \hat{a}^\dagger)$.

Therefore $\langle X \rangle_\alpha = \sqrt{\frac{\hbar}{2m\omega}} (\langle \hat{a} \rangle_\alpha + \langle \hat{a}^\dagger \rangle_\alpha) = \sqrt{\frac{\hbar}{2m\omega}} (\alpha + \alpha^*) = \sqrt{\frac{2\hbar}{m\omega}} \Re(\alpha)$.

Similarly, $\langle P \rangle_\alpha = -i\sqrt{\frac{m\hbar\omega}{2}} (\langle \hat{a} \rangle_\alpha - \langle \hat{a}^\dagger \rangle_\alpha) = -i\sqrt{\frac{m\hbar\omega}{2}} (\alpha - \alpha^*) = \sqrt{2m\hbar\omega} \Im(\alpha)$.

Now for the variances. First, $X^2 = \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2)$.

Using $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ and the coherent-state expectation values

$$\langle \hat{a} \rangle = \alpha, \quad \langle \hat{a}^\dagger \rangle = \alpha^*, \quad \langle \hat{a}^2 \rangle = \alpha^2, \quad \langle (\hat{a}^\dagger)^2 \rangle = (\alpha^*)^2, \quad \langle \hat{a}^\dagger \hat{a} \rangle = |\alpha|^2,$$

$$\text{we find } \langle X^2 \rangle_\alpha = \frac{\hbar}{2m\omega} (\alpha^2 + (\alpha^*)^2 + |\alpha|^2 + (|\alpha|^2 + 1)) = \frac{\hbar}{2m\omega} (\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2 + 1).$$

$$\text{On the other hand, } \langle X \rangle_\alpha^2 = \frac{\hbar}{2m\omega} (\alpha + \alpha^*)^2 = \frac{\hbar}{2m\omega} (\alpha^2 + (\alpha^*)^2 + 2|\alpha|^2). \text{ Therefore } \Delta X_\alpha^2 = \langle X^2 \rangle_\alpha - \langle X \rangle_\alpha^2 = \frac{\hbar}{2m\omega},$$

$$\text{so } \Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}.$$

$$\text{Likewise one finds } \Delta P_\alpha = \sqrt{\frac{m\hbar\omega}{2}}.$$

So $\Delta X_\alpha \Delta P_\alpha = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}$. So coherent states are minimum-uncertainty states.

- (g) Derive the position-space form of the coherent state.

Calculation

One elegant way is to solve $(\hat{a} - \alpha)\phi_\alpha(x) = 0$ in the position representation. In x -space, $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}$. Therefore the equation becomes $\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx} - \alpha\right)\phi_\alpha(x) = 0$. Multiply by $\sqrt{\frac{2m\omega}{\hbar}}$: $\left(\frac{m\omega}{\hbar}x + \frac{d}{dx} - \sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\phi_\alpha(x) = 0$. So $\frac{d\phi_\alpha}{dx} = \left(\sqrt{\frac{2m\omega}{\hbar}}\alpha - \frac{m\omega}{\hbar}x\right)\phi_\alpha$. Integrating, $\phi_\alpha(x) = C \exp\left(\sqrt{\frac{2m\omega}{\hbar}}\alpha x - \frac{m\omega}{2\hbar}x^2\right)$. Writing $\alpha = \Re(\alpha) + i\Im(\alpha)$ and completing the square gives the Gaussian wave packet form $\phi_\alpha(x) = e^{i\theta_\alpha} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{(x - \langle X \rangle_\alpha)^2}{4(\Delta X_\alpha)^2} + \frac{i}{\hbar}\langle P \rangle_\alpha x\right]$, with $\langle X \rangle_\alpha = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha)$, $\langle P \rangle_\alpha = \sqrt{2m\hbar\omega}\Im(\alpha)$, $\Delta X_\alpha = \sqrt{\frac{\hbar}{2m\omega}}$. Therefore $|\phi_\alpha(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left[-\frac{(x - \langle X \rangle_\alpha)^2}{2(\Delta X_\alpha)^2}\right]$. This is a Gaussian wave packet centered at $\langle X \rangle_\alpha$ with width ΔX_α .

Bonus — time evolution of coherent states At time $t = 0$ the oscillator is prepared in the coherent state $\psi(x,0) = \phi_{\alpha_0}(x)$. Study the time evolution under $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$.

- (a) Show that $\psi_t = e^{-i\hat{H}t/\hbar}\psi_0 = e^{-i\omega t/2}\phi_{\alpha_t}$, $\alpha_t = \alpha_0 e^{-i\omega t}$.

Calculation

Expand the initial state in the energy eigenbasis: $\phi_{\alpha_0} = e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} \phi_n$. The time-evolution operator acts as $e^{-i\hat{H}t/\hbar}\phi_n = e^{-i\omega(n+1/2)t}\phi_n$. Therefore $\psi_t = e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\omega(n+1/2)t} \phi_n = e^{-i\omega t/2} e^{-|\alpha_0|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha_0 e^{-i\omega t})^n}{\sqrt{n!}} \phi_n$. Since $|\alpha_0 e^{-i\omega t}| = |\alpha_0|$, this is exactly $\psi_t = e^{-i\omega t/2} \phi_{\alpha_t}$, $\alpha_t = \alpha_0 e^{-i\omega t}$.

- (b) Compute the expectation values of position, momentum, and energy, and compare them to classical mechanics.

Calculation

Since at time t the state is again a coherent state with parameter $\alpha_t = \alpha_0 e^{-i\omega t}$, the formulas from the previous section apply directly: $\langle X \rangle_t = \sqrt{\frac{2\hbar}{m\omega}}\Re(\alpha_t)$, $\langle P \rangle_t = \sqrt{2m\hbar\omega}\Im(\alpha_t)$. Write $\alpha_0 = |\alpha_0|e^{i\phi}$. Then $\alpha_t = |\alpha_0|e^{i(\phi - \omega t)}$. So $\langle X \rangle_t = \sqrt{\frac{2\hbar}{m\omega}}|\alpha_0|\cos(\omega t - \phi)$ and $\langle P \rangle_t = -\sqrt{2m\hbar\omega}|\alpha_0|\sin(\omega t - \phi)$. These are exactly the classical solutions for a harmonic oscillator: $x_{\text{cl}}(t) = A\cos(\omega t - \phi)$, $p_{\text{cl}}(t) = m\dot{x}_{\text{cl}}(t)$, with amplitude $A = \sqrt{\frac{2\hbar}{m\omega}}|\alpha_0|$. The energy expectation value is $\langle H \rangle_t = \hbar\omega(|\alpha_t|^2 + \frac{1}{2})$. Since $|\alpha_t| = |\alpha_0|$, this is constant: $\langle H \rangle_t = \hbar\omega(|\alpha_0|^2 + \frac{1}{2})$. The classical harmonic oscillator energy is also conserved. So the expectation values of X and P follow the classical trajectory exactly, while the energy remains constant.

(c) Compute the uncertainties ΔX_t and ΔP_t .

Solution

Since ψ_t is again a coherent state for all t , the uncertainties remain those of any coherent state: $\Delta X_t = \sqrt{\frac{\hbar}{2m\omega}}$, $\Delta P_t = \sqrt{\frac{m\hbar\omega}{2}}$. Therefore $\Delta X_t \Delta P_t = \frac{\hbar}{2}$ at all times.

So the wave packet oscillates without changing shape: its center moves classically, while its width stays constant.

Additional Notes / Side Calculations

Side Calculation

identities for the harmonic oscillator: $\hat{a}\phi_n = \sqrt{n}\phi_{n-1}$, $\hat{a}^\dagger\phi_n = \sqrt{n+1}\phi_{n+1}$, $\hat{N} = \hat{a}^\dagger\hat{a}$, $X = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$, $P = -i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger)$.

Gaussian integral: $\frac{1}{\pi} \int_{\mathbb{C}} d^2\alpha e^{-|\alpha|^2} \alpha^n (\alpha^*)^m = n! \delta_{nm}$.

Notes

Physical interpretation:

- The Bernoulli distribution models a binary random event; its fluctuations are maximal at $p = 1/2$.
- For the Poisson distribution, mean and variance coincide, which is a characteristic signature of counting statistics.
- The Gaussian has vanishing odd central moments because it is symmetric about its mean.
- In a constant electric field, the harmonic oscillator potential is merely shifted in space; the shape and level spacing are unchanged.
- Coherent states are the most classical oscillator states: the packet center follows the classical orbit, while the quantum uncertainty stays minimal and constant.